Infosys Springboard Virtual Internship Report

Project Title: Infosys_HotelRevAI – Revenue Analysis for Hotels

1. Introduction

The hospitality industry generates large amounts of booking and revenue-related data daily. Analyzing this data effectively helps hotels optimize occupancy, pricing, and overall revenue performance.

As part of the Infosys Springboard Virtual Internship, this project focuses on performing revenue analysis using hotel booking data. The main objective was to clean raw data, structure it properly, and derive meaningful business insights using analytical tools.

2. Objective of the Project

- To collect a relevant hotel booking dataset
- To clean and preprocess the dataset for analysis
- To perform revenue-related analysis
- To create calculated measures using DAX functions
- To generate business insights from structured data

3. Dataset Description

The dataset was collected from Kaggle and contains historical hotel booking data including booking dates, arrival details, number of guests, length of stay, ADR (Average Daily Rate), market segment, distribution channel, cancellation status, country, and customer type.

This dataset provided sufficient information to perform revenue and booking pattern analysis.

4. Data Collection

The dataset was downloaded in CSV format from Kaggle and imported into Power BI for processing. Before analysis, the dataset contained missing values, redundant columns, and inconsistent formatting, which required cleaning and transformation.

5. Data Cleaning & Transformation (Using Power Query)

Data preprocessing was performed using Power Query Editor. The following steps were taken:

- Removed unnecessary columns
- Handled missing values
- Removed duplicate records
- Corrected data types (dates, numeric, categorical)
- Standardized categorical values
- Created new calculated columns:
  - Total Stay = Weekend Nights + Weekday Nights
  - Total Guests = Adults + Children + Babies
  - Revenue = ADR × Total Stay

After cleaning, the dataset became structured and analysis-ready.

6. Data Analysis Using DAX

After cleaning the data, DAX (Data Analysis Expressions) functions were used in Power BI to create calculated measures.

Key DAX Measures Created:

```
Total Revenue = SUMX(Data, Data[ADR] * Data[Total Stay])

Total Bookings = COUNT(Data[Booking ID])

Cancellation Rate = DIVIDE(CALCULATE(COUNT(Data[Booking ID], Data[is_canceled]=1)),
COUNT(Data[Booking ID]))

Average ADR = AVERAGE(Data[ADR])
```

7. Insights Derived

- Revenue trends showed seasonal variation
- Certain market segments contributed higher revenue
- Higher cancellation rates reduced total revenue
- Distribution channels influenced booking volume
- Longer stays generated higher revenue

8. Tools & Technologies Used

- Kaggle – Dataset Source
- Power BI
- Power Query – Data Cleaning
- DAX – Revenue Analysis

9. Learning Outcomes

- Practical experience in data collection and preprocessing
- Hands-on experience in Power Query
- Strong understanding of DAX functions
- Ability to create revenue-based analytical measures
- Experience in converting raw data into business insights

10. Conclusion

The Infosys_HotelRevAI project focused on transforming raw hotel booking data into structured insights using Power Query and DAX functions. By cleaning the dataset and creating analytical measures, meaningful revenue insights were generated to support data-driven decision-making in hotel revenue management.